System Skills A03 Mysterious Cypher

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1. Determine the Key

The key is computed in the first few lines of the main routine before the decryption loop begins:

- **movl num1(%rip), %eax**: This instruction loads the value num1 from the data section (which is **2**) into the 32-bit %eax register.
- **movl num2(%rip), %ecx**: This loads the value num2 (**-5**) into the %ecx register.
- **addl %ecx, %eax**: This adds the value in %ecx to %eax. 2 + (-5) = -3.
- **movb %al, %bl**: This takes the lowest 8 bits of %eax (which is %al) and stores it in %bl. %bl now holds our decryption key.

Final Key Value: The computed decryption key is **-3**.

2. Recover the Plaintext

The ciphertext provided in the data section is ddbw. The program processes this string character by character in the decrypt_loop.

The core logic happens at **subb %bl, %al**:

- %al holds the current ASCII character from the ciphertext.
- %bl holds our key (-3).
- The instruction subtracts %bl from %al. Since it's subtracting a negative number the operation is **ASCII value + 3**.

This then applies a +3 shift to each character in ddbw:

- **d** (ASCII 100) + 3 = 103 → **g**
- **d** (ASCII 100) + 3 = 103 → **g**
- **b** (ASCII 98) + 3 = 101 → **e**
- **w** (ASCII 119) + 3 = 122 → **z**

Original Plaintext Message: The decrypted string is **ggez**.

What I Discovered:

How is %al, %rax and %eax are all connected
When analyzing between **main** and **decrypt_loop** I was initially confused on why in **main**, **%al** at the end was -3, yet in **decrypt_loop** it acts as the individual letters in the encrypted message. After looking through my notes and previous slides, I eventually realized that **%al**, **%eax**, and **%rax** are all connected. **%rax** is the whole register, **%eax** is the bottom half of the register and **%al** is the bottom 8 bits of the register. By doing **movl num1(%rip), %eax** and **movzbq (%rdi), %rax** we are affecting **%al** too.

How the decrypting process works
This also explains how **%al** is used in decrypting the message. After the message, **(%rdi)**, gets put into **%rax**, **%al** is also affected by this and is representing the lower 8 bits of **%rax**: the first letter in the encrypted message. By doing **inc %rdi** we essentially move where **%al** represents, thus moving from one character to the next, making decrypting each letter using **subb %bl, %al** work.